Coastal Mangrove Restoration - Field Report, Season Three

Site and methods

This report describes the third planting season of a community mangrove restoration project on a degraded mudflat near the mouth of a small tidal creek. The site had lost most of its mangrove cover over three decades, first to fuelwood cutting and then to shallow aquaculture ponds that were abandoned when yields fell. By the start of the project only scattered grey mangrove shrubs remained along the upper edge of the flat, and the lower mud was bare at low tide.

The project is run by a fishers' cooperative from the neighbouring village with technical support from a university marine biology department. Village members collect propagules, raise seedlings in a shaded nursery, plant at low tide and record survival every quarter. The university team analyses the records, measures sediment and water quality, and trains a small group of village monitors so that the work can continue without outside staff.

Before planting, the team surveyed the elevation of the flat at fifty points using a simple water-level method. Mangrove species differ in how long they tolerate being underwater, and earlier projects in the region failed largely because seedlings were planted too low, where they stayed submerged for most of each tide. The survey divided the site into three zones by how many hours each spot remained flooded during a typical spring tide.

In the first two seasons the team also reopened two blocked tidal channels so that water could drain the inner flat properly. Stagnant water had left the sediment saline and low in oxygen, and seedlings planted there in a pilot trial had died within weeks. After the channels were cleared, salinity in the inner zone fell steadily over the following year.

In season three the team planted 18,400 seedlings across the three zones, of which 61 percent were grey mangrove and the remainder red and black mangrove, chosen according to the flooding duration of each zone.

Survival and growth

Survival was measured by counting live seedlings in fixed plots of ten metres by ten metres, twelve plots in each zone, at three, six and twelve months after planting. A seedling was counted as alive if it had at least one green leaf. Plots were marked with bamboo poles painted at the top so that monitors could find them at high tide, and each plot was photographed from the same pole at every visit.

Survival in the upper and middle zones improved sharply compared with the first season. Monitors attributed this mainly to better species matching and to planting only during the cooler months, when seedlings lose less water through their leaves. In the lowest zone survival remained poor, and the team now believes that part of this zone sits too low for any mangrove species and should be left as open mudflat for wading birds.

Barnacles and crabs caused the most damage to young seedlings. Barnacles settled on stems in the lowest zone and weighed them down, while crabs cut the stems of fresh seedlings in the middle zone during the first weeks. The village monitors tried sleeves made from cut plastic bottles around the stems, but abandoned them after one season because the sleeves trapped silt and some ended up as litter on the shore.

Height growth was slower than expected in the first year but picked up in the second, once roots reached the firmer sediment below the surface layer. The tallest grey mangroves in the upper zone now exceed a metre and have begun to produce flowers, which the monitors record because local seed supply will reduce the cost of future planting.

Twelve months after planting, survival stood at 74 percent in the upper zone, 58 percent in the middle zone and only 19 percent in the lowest zone.

Community and costs

The project depends on unpaid and paid village labour. Propagule collection and nursery care are paid at the local daily wage, while planting days are organised as community events with a shared meal. Women's self-help groups run two of the three nurseries and keep the nursery accounts, which the cooperative audits every six months together with a volunteer accountant from the nearby town.

The total cost of season three, including nursery materials, wages, transport and monitoring equipment, was lower per surviving seedling than in either of the earlier seasons. Most of the saving came from producing seedlings locally rather than buying them from a government nursery two hours away, and from fewer seedlings dying after planting, which meant fewer had to be replaced.

Fishers report more juvenile crabs and small fish in the reopened channels, although the project has not yet carried out a formal fish survey, so these observations should be treated as anecdotal. The university team has applied for a separate grant to run a structured survey with nets of standard mesh size at fixed points across a full year.

The cooperative has agreed with the forest department that restored areas will not be cut, and that the village will patrol the site during the dry season when fuelwood collection is most common. Two monitors also record any new aquaculture bunds being built nearby, since a single new pond can block the channels again.

The project's next goal is to hand monitoring entirely to the village by the end of season five, after which the university will visit only once a year to check plot records and repeat the elevation survey.